10. Monday, April 21. Order continued. Divisibility (§3)

Last Time: Order axioms for $\mathbb{R}$: There Exists subset $\mathbb{R}^+ \subseteq \mathbb{R}$ satisfying:

12) Positive closure ( Closed under + and · )

13) Trichotomy.

Elements in $\mathbb{R}^+$ are called positive. Denote $\mathbb{Z}^+ := \mathbb{Z} \cap \mathbb{R}^+$ and $\mathbb{Q}^+ := \mathbb{Q} \cap \mathbb{R}^+$

**Definition 1**: For real numbers $a, b$,

- $a < b$ means that $b - a \in \mathbb{R}^+$
- $a \leq b$ means $a < b$ or $a = b$ $\leq$ : \leq
- $a > b$ means $a - b$ is positive (i.e. $b < a$)
- $a \geq b$ means $a > b$ or $a = b$ $\geq$ : \geq
- $a$ is negative means $-a$ is positive.

**Theorem 2**: Let $a \in \mathbb{R}$. Then $a \in \mathbb{R}^+$ if and only if $a > 0$.

**Technique**: To prove a biconditional $P \longleftrightarrow Q$, prove $P \longrightarrow Q$ and $Q \longrightarrow P$

We will use **Lemma 3**. We have $-0 = 0$.

**Proof:** Since $0$ is the additive identity, $0 + 0 = 0$.
But this equation also says that $0$ is an additive inverse of $0$.
By Theorem 9.3(c), additive inverses are unique, so $0 = -0$. □

**Proof** of Theorem 2: Fix $a \in \mathbb{R}$.

[$\Rightarrow$] (\Rightarrow) Suppose $a \in \mathbb{R}^+$. We want to show that $a > 0$, in other words, that $a - 0 \in \mathbb{R}^+$.

So we compute

$$a - 0 = a + (-0) = a + 0 = a$$

(Definition; Lemma 6; Additive id.)

Need $-0$. So let's write a lemma.

Since we know that $a \in \mathbb{R}^+$ and $a - 0 = a$, we also get $a - 0 \in \mathbb{R}^+$ by substitution so $a > 0$ as desired.

[$\Leftarrow$] (\Leftarrow) Suppose now $a > 0$, and we want to show that $a \in \mathbb{R}^+$.

By definition, $a > 0$ means $a - 0 \in \mathbb{R}^+$. But again,

$$a - 0 = a + (-0) = a + 0 = a$$

Since $a - 0 \in \mathbb{R}^+$ and $a = a - 0$, we get $a \in \mathbb{R}^+$, as desired. □

**Theorem 4**: 1 is positive.

**Technique**: (Proof by contradiction) Assume the Theorem is False. Deduce a contradiction. Therefore your assumption is wrong, and the Theorem is True.

**Proof of Theorem 4**: Suppose for a contradiction that 1 is not positive.

By Trichotomy, either $-1$ is positive or $1=0$.

Case 1: $-1$ is positive. By closure of $\mathbb{R}^+$ we get that $(-1)(-1) \in \mathbb{R}^+$

But by Theorem 9.4, $(-1)(-1) = 1$, so we get $1 \in \mathbb{R}^+$, This is a contradiction!

Case 2: $1=0$. This immediately contradicts axiom 5, which states that $1 \neq 0$.

Either case gives a contradiction, so 1 must be positive $\square$

## Well-Ordering Axiom (for positive integers)

**Definition 5:** Let $S$ be a set of positive integers. A **minimum element** for $S$ is an element $a \in S$ such that $a \leq b$ for all $b \in S$.

**Axiom 14) Well-Ordering Property.**
If $S$ is a non-empty set of positive integers, then $S$ has a minimum element.

**Note:** Not true in $\mathbb{R}^+$: open interval $(1,2)$ has no minimal element.

**Theorem 6:** Let $a \in \mathbb{R}$. If $0 < a < 1$, then $0 < a^2 < a$ **Proof:** HW 4 □

**Theorem 7:** The integer 1 is the smallest positive integer.

**Proof:** Suppose for a contradiction there is some $x \in \mathbb{Z}$ with $0 < x < 1$.
Then the set $S = \{x \in \mathbb{Z} \mid 0 < x \text{ and } x < 1\}$ is non-empty.

Since $S \subseteq \mathbb{Z}^+$ is non-empty, by the well-ordering property, it has a minimum element, say $y$.
By closure of $\mathbb{Z}$, we have $y^2 \in \mathbb{Z}$. And by Thm 3 above, since $0 < y < 1$, we have $0 < y^2 < y < 1$. Therefore $y^2 \in S$ and $y^2 < y$, which contradicts that $y$ is minimum.
□

**Technique** (Infinite Descent) To prove that something in $\mathbb{Z}^+$ does not exist:

1) Assume for a contradiction the thing exists and take the minimum one
2) Find a smaller one. This is a contradiction.

## Divisibility

**Definition 5:** We say that a non-zero integer $m$ **divides** an integer $n$ if there exists some integer $q$ so that $n = m \cdot q$

In this case, we say that $m$ is a **divisor** or a factor of $n$.

and $n$ is a **multiple** of $m$.

The integer 0 is **not** a divisor of any integer.

**Example:** 7 divides 21, because $21 = 7 \cdot 3$. So 7 is a divisor of 21, and 21 is a multiple of 7.

**Notation:** If $a$ divides $b$ we write $a \mid b$ (\mid)

**Theorem 6:** For every nonzero integer $a$, we have $a \mid a$.

**Proof:** Let $a$ be a nonzero integer. By the identity axiom, $a = a \cdot 1$, and since axiom 5 holds for $\mathbb{Z}$, we have $1 \in \mathbb{Z}$. Hence $a \mid a$ by definition of divisibility.

**Theorem 7**: Let $a,b,c \in \mathbb{Z}$ with $a \neq 0$ and $b \neq 0$. If $a \mid b$ and $b \mid c$, Then $a \mid c$.

**Proof**: Fix $a,b,c \in \mathbb{Z}$ with $a \neq 0$ and $b \neq 0$. Suppose $a \mid b$ and $b \mid c$.

By definition, this means There exist $q, p \in \mathbb{Z}$ such That $b = a \cdot q$ and $c = b \cdot p$.

Want $c = b \cdot k$ some $k$

By substitution, $c = bp = (a \cdot q) \cdot p$. By associativity $c = a \cdot (q \cdot p)$.

But $q \cdot p \in \mathbb{Z}$, because $\mathbb{Z}$ is closed.

Hence $a \mid c$ by definition of divisibility. □

## Evens and odds

Notation: $2 = 1+1$. So 2 is a positive integer different from 1.

**Recall**: An integer $n$ is:
- Even if $n = 2k$ for some $k \in \mathbb{Z}$. i.e, $2 \mid n$
- Odd if $n = 2k+1$ for some $k \in \mathbb{Z}$.

**Theorem 8**: The integer 1 is odd

**Proof**: This is clear since $1 = 2 \cdot 0 + 1$.

**Theorem 9**: The integer 1 is not even.

**Proof**: Suppose for a contradiction That There exists $k \in \mathbb{Z}$ such That $1 = 2 \cdot k$.

We consider four cases That exhaustall possibilities, and show each leads to a contrad

**Case 1**: $k$ is negative. If $k$ is negative, Then $(-k)$ is positive, so $2 \cdot (-k)$ is positive by positive Closure. Since $2(-k) = -2k$, This means That $2k$ is negative. But $2k = 1$ is positive

By Thm 1, so this is a contradiction by Trichotomy.

Case 2: $K = 0$. If $K = 0$, then $1 = 2 \cdot K = 2 \cdot 0 = 0$. So $1 = 0$, which contradicts the identity ax

Case 3: $K = 1$. If $K = 1$, then $1 = 2 \cdot K = 2 \cdot 1 = 2 = 1 + 1$.

By cancellation law, we get $0 = 1$, which is a contradiction.

Case 4: $K$ is positive but $K \neq 1$. Since 1 is the smallest positive integer, we have $K > 1$.

By a HW exercise, since 2 is positive we get $2 \cdot K > 2 \cdot 1$, ie: $2K > 2$, Hence $1 > 2$.

This contradicts 1 is the smallest positive integer.

Any possibility for $K$ leads to a contradiction, so it cannot exist, as desired □

Technique (Breaking into cases) To prove something is true, prove that it is true in several specific cases, as long as these cases exhaust all possibilities.